Machine Learning for Electricity Demand Forecasting:
A Comparison of Five Models Using 2025 CAISO Data

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SUMMARY

Electricity demand changes throughout the day and across different seasons, so being able to predict future demand can help us better understand how electricity grids need to operate. In this project, we wanted to find out whether machine-learning models could predict electricity demand 24 hours ahead better than simply using the demand from the previous day.

We used hourly California electricity demand data from 2025 together with historical weather data from Open-Meteo. After cleaning the data and creating additional time-based features, we had 8,544 hourly observations to work with. We compared five approaches: a simple 24-hour persistence method, Ridge Regression, Random Forest, Gradient Boosting, and a Long Short-Term Memory (LSTM) model. We tested the models on the later part of the year so that the models were trained using earlier data and then tested on data they had not seen before. We measured their performance using mean absolute error (MAE), root mean squared error (RMSE), and mean absolute percentage error (MAPE).

Gradient Boosting gave the best results among the five machine-learning approaches. It had an MAE of 717 MW and a MAPE of 2.91%, compared with 1,049 MW and 4.28% for the simple persistence method. This represents a 31.7% reduction in MAE. We also found that electricity demand has a strong daily pattern: demand at a given hour was strongly related to demand at the same hour on the previous day, with a 24-hour correlation of 0.913.

Our results show that machine-learning models, especially tree-based models such as Gradient Boosting, can improve short-term electricity demand forecasts compared with a simple previous-day estimate. At the same time, the strong daily pattern in electricity demand shows why simple forecasting methods can still provide a useful starting point. More testing using data from multiple years, regions, and actual weather forecasts would be needed before deciding how well these models could work in a real electricity grid.

1. Introduction

Electricity grids are complex systems in which generation and consumption must remain closely balanced. Because electricity demand changes by hour, day, season, and weather conditions, grid operators need forecasts to make decisions about generation, reserves, and other operating resources. A useful forecast can help operators prepare for expected demand while reducing the need to rely on last-minute adjustments.

Electricity demand forecasting can be approached using statistical models, operational forecasting methods, and machine-learning techniques. A particularly important benchmark is a persistence forecast: if demand follows strong daily patterns, tomorrow's demand at a given hour may be similar to today's demand at the same hour. A machine-learning model should therefore demonstrate improvement over this simple benchmark before its additional complexity is considered useful [6].

This study asks: Can machine learning predict electricity demand more accurately than a simple 24-hour persistence baseline?

We had three goals: (1) look at how demand changes by hour, by season, and with the weather; (2) compare five forecasting approaches using the same tests; and (3) find out which information the best model relied on most.

2. Background

2.1 Electricity Demand and the Grid

Electricity is generated from multiple sources and delivered to consumers through transmission and distribution systems. Because large-scale electricity storage remains limited relative to total grid demand, grid operators must continuously balance supply and demand. Forecasting demand ahead of time can help operators prepare generation and other resources.

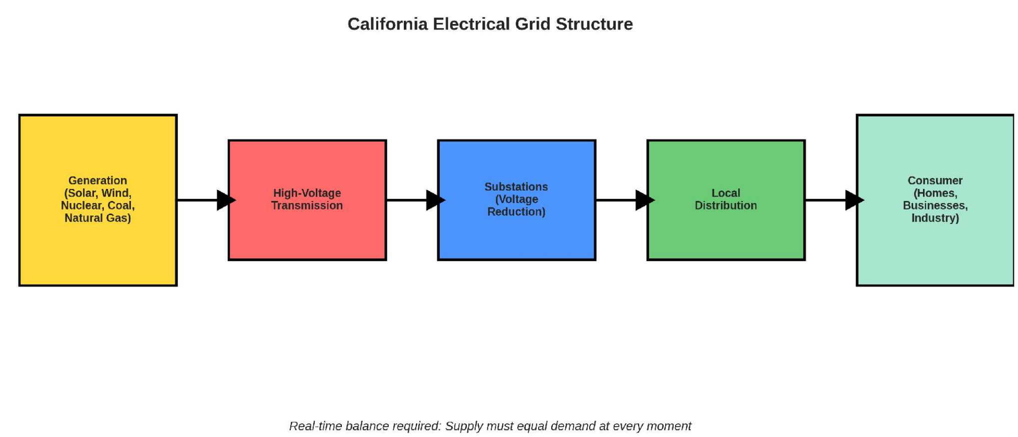

Figure 1. Simplified California electrical grid structure from generation through transmission, distribution, and consumers.

2.2 Machine Learning for Demand Forecasting

Machine learning identifies patterns in historical data and uses those patterns to make predictions on new observations. For electricity demand forecasting, useful inputs can include recent demand, time-of-day and day-of-week patterns, weather variables, holidays, and rolling statistics. This study focuses on supervised learning, where historical inputs are paired with known future demand values during model development.

3. Data and Exploratory Analysis

3.1 Data Sources and Dataset Construction

We used hourly California electricity demand data for 2025 [1] and matched it with historical weather data from Open-Meteo. The weather data included temperature, humidity, dewpoint, and wind speed for the Sacramento area [2]. We started with 8,760 hourly observations for the year. After removing rows that could not be used because they needed earlier or later data, we had 8,544 observations for the analysis.

Demand ranged from 15,983 MW to 43,860 MW, with an average of 25,621 MW. The highest demand was 43,860 MW on August 21 at about 8:00 PM. We also found one unusual reading on July 31 at 9:00 PM UTC. It was 11,819 MW while the surrounding values were close to 30,000 MW, so we replaced it with an estimate of 29,883.5 MW based on the values around it. After cleaning the data, there were no missing values or missing hourly timestamps.

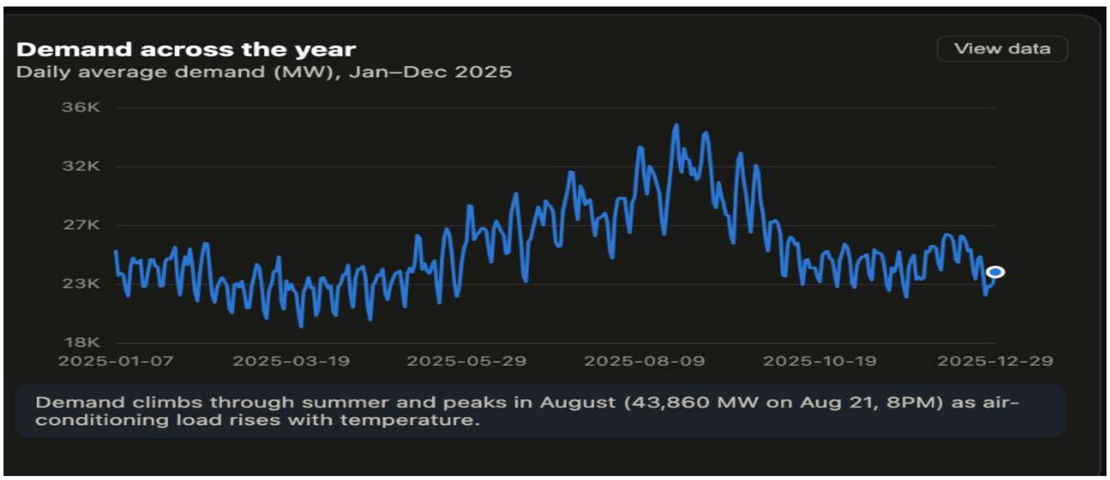

Figure 2. Daily average California electricity demand during 2025. The underlying hourly dataset reached a peak of 43,860 MW on August 21 at approximately 8:00 PM.

3.2 Exploratory Findings

We looked at the data before building the forecasting models to see what patterns were present. One clear pattern was seasonal. Average demand was 22,332 MW in March and 31,020 MW in August, about 39% higher in August. August also had more variation from hour to hour than March.

We also found that demand was strongly related to what happened earlier. Demand one hour earlier had a correlation of 0.966, while demand at the same hour the previous day had a correlation of 0.913. This helps explain why using the previous day's demand is already a useful starting point for a forecast.

Temperature was related to demand, and weekday peak demand was about 1,980 MW higher than weekend peak demand. These patterns helped us decide which time and weather information to give the models.

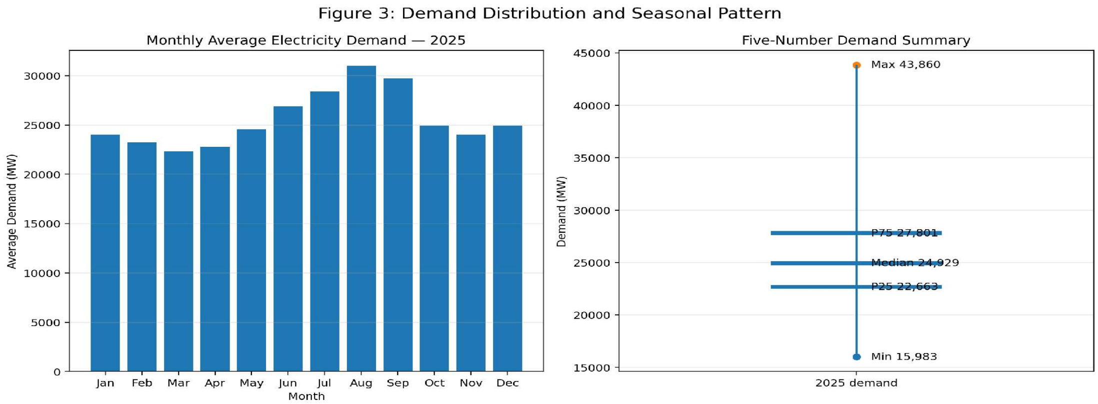

Figure 3. Monthly average electricity demand and five-number summary of the 2025 hourly demand dataset.

3.3 Data Quality and Preprocessing

- After cleaning the data, we checked it again before training the models. The final dataset had 8,544 usable hourly observations, no missing values, and no missing hourly timestamps. The one unusual demand reading was corrected using the values immediately before and after it.

4. Methodology

4.1 Feature Engineering

We built features to capture the patterns we found when we explored the data. These included recent demand values, demand from 24 hours and further back, rolling averages and standard deviations, hour and day-of-week encoded so that the end of one day connects to the start of the next, holiday and season flags, and the weather variables.

- Lag features: previous-hour demand, same-hour demand from the previous day, and longer historical lags.
- Rolling statistics: 24-hour and 7-day rolling averages and standard deviations.
- Cyclical time features: sine/cosine representations of hour of day and day of week.
- Calendar features: holiday indicators and seasonal flags.
- Forecast target: electricity demand 24 hours after the forecast origin.

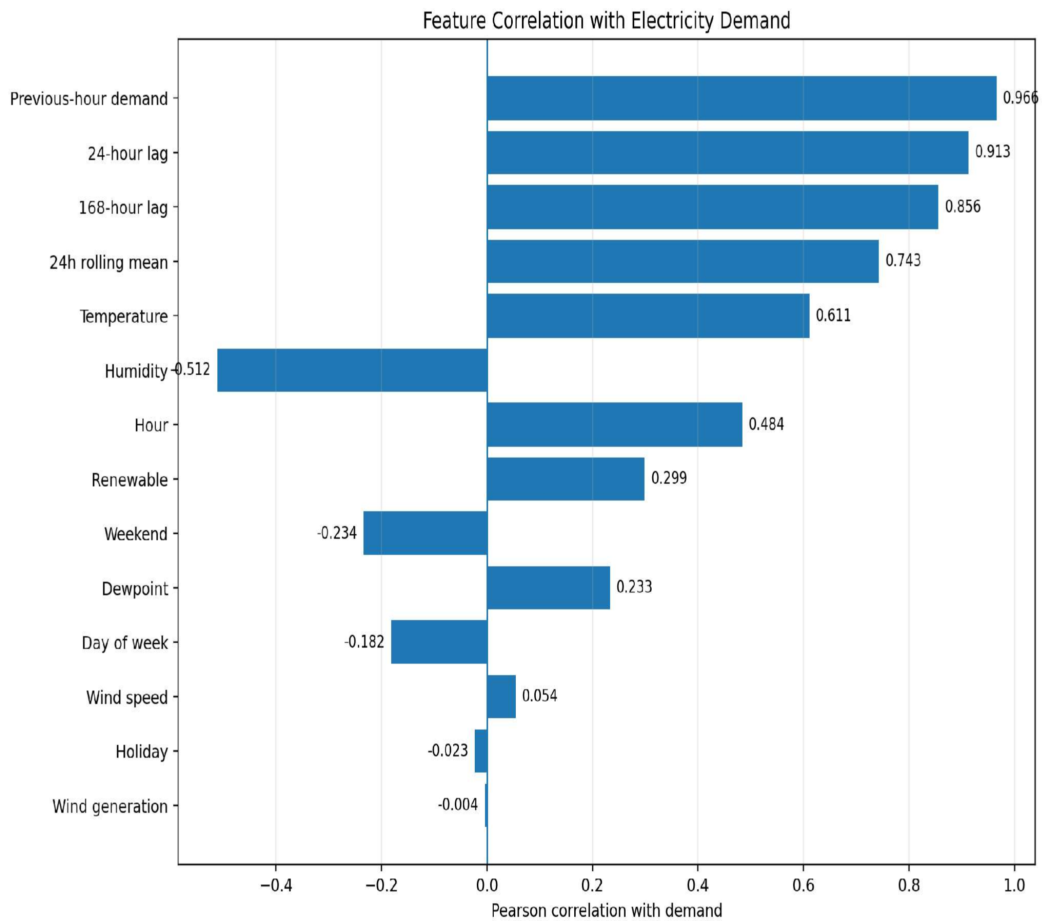

Figure 4. Pearson correlation of selected demand, weather, renewable, and time features with electricity demand.

4.2 Forecasting Design and Leakage Prevention

We wanted the model to predict demand 24 hours ahead: for a forecast made at time t, the model predicts demand at t+24. We split the data by date instead of randomly, so that data from the future could never end up in the training set. Training used 7,103 rows and testing used 1,441 rows held back until the end. The test period runs from October 30 through December 29, 2025, which is 1,441 hourly readings across 60 days once both end hours are counted. Splitting this way keeps the forecasting problem in the right time order.

We lined up the weather data so that no weather reading from inside the forecast window was used to predict that window. That way the model never sees information it would not have had at the time. One thing to be clear about, though: because we were testing on past data, we used the weather that actually happened. A real grid operator would only have a weather forecast, which is not perfect. This is an important limit on our results.

4.3 Evaluation Metrics

- MAE (Mean Absolute Error): average absolute difference between predicted and actual demand, measured in MW.
- RMSE (Root Mean Squared Error): gives greater weight to larger forecasting errors.
- MAPE (Mean Absolute Percentage Error): expresses average absolute error as a percentage of actual demand.

4.4 Models Compared

We compared five approaches. The persistence method is the simplest possible starting point, since it does no learning at all. Ridge Regression is a straight-line model for comparison, Random Forest and Gradient Boosting are tree-based models that can pick up patterns that are not straight lines, and the LSTM is a neural network that looks back over a week of data.

- 24-hour persistence baseline: forecast demand at t+24 using demand observed 24 hours earlier at the same hour.
- Ridge Regression: a regularized linear model using the engineered features, implemented with scikit-learn Ridge and alpha = 10.0 [9]. We used Ridge rather than plain linear regression because our lag features overlap a lot, and Ridge is built to handle inputs that are closely related to each other [3].
- Random Forest: an ensemble of decision trees that can capture nonlinear relationships and feature interactions [4].
- Gradient Boosting: sequentially builds decision trees so later trees focus on errors made by earlier trees [8].
- LSTM: a recurrent neural network designed to learn temporal patterns from sequences, using a 168-hour lookback [5].

4.5 Reproducibility and Code Availability

The project notebooks used for dataset construction and model training are publicly available in the project repository [7].

5. Results

5.1 Comparative Performance

Below are our results on the test period we held back. Every model was measured the same way, using MAE, RMSE, and MAPE.

Table 1. Model performance on the held-out 60-day test set.

Model	MAE (MW)	RMSE (MW)	MAPE (%)	Change vs. Baseline
24-hour Persistence	1,049	1,451	4.28%	—
Ridge Regression	1,698	2,030	6.82%	61.8% worse
Random Forest	727	1,024	2.96%	30.8% lower MAE
Gradient Boosting	717	980	2.91%	31.7% lower MAE
LSTM	1,064	1,418	4.37%	1.4% worse

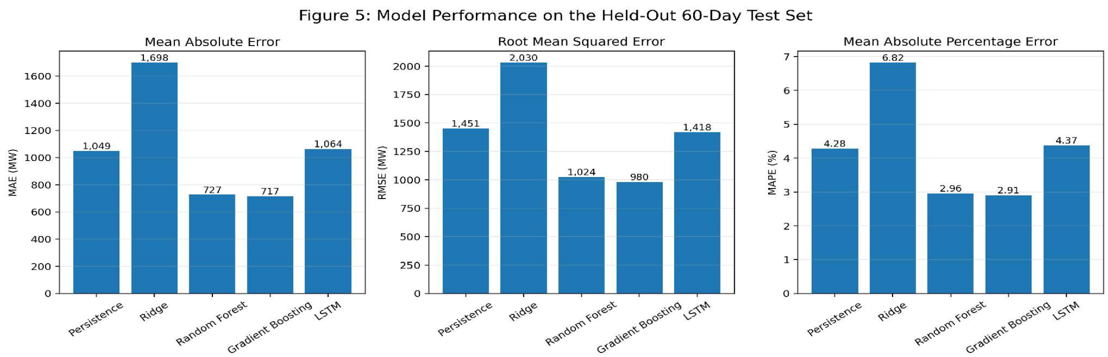

Figure 5. Model performance comparison on the held-out test period.

Table 2. Key model settings used in the five forecasting models.

Model	Library / method	Key settings
24-hour Persistence	Custom baseline	24-hour lag
Ridge Regression	scikit-learn	alpha = 10.0
Random Forest	scikit-learn	300 trees; max depth 14; seed 42
Gradient Boosting	scikit-learn	300 trees; max depth 3; learning rate 0.05; seed 42
LSTM	PyTorch	168-hour lookback; 2 layers × 64 hidden; dropout 0.2; batch 64; max 40 rounds; patience 5; seed 42

5.2 Key Findings

Gradient Boosting produced the lowest error among the five tested approaches. Its MAE was 717 MW, compared with 1,049 MW for the persistence baseline, representing a 31.7% reduction in MAE. Random Forest produced a very similar result, with an MAE of 727 MW and a 30.8% reduction relative to the baseline.

The persistence baseline was surprisingly strong. Its MAPE of 4.28% reflects the strong daily and weekly patterns observed in the exploratory analysis. The fact that Gradient Boosting improved substantially over this baseline suggests that the model was able to use additional nonlinear information beyond simple persistence.

Ridge Regression did worse than the persistence baseline. Our lag features overlap a lot, so the inputs are closely related to each other, and Ridge is designed to cope with that. Even so, a straight-line relationship did not capture the demand patterns as well as the tree-based models did in this experiment.

The LSTM had a MAPE of 4.37%, roughly tied with the persistence baseline. It used a one-week lookback and stopped training early once it stopped improving on a validation slice. So the more complicated sequence model did not beat the tree models here. That does not mean LSTMs are worse in general, only that ours was not better on this problem.

5.3 Feature Importance

Feature-importance analysis for the Gradient Boosting model showed that recent demand was the dominant model input, accounting for approximately 85.6% of the reported importance. Hour-of-day and day-of-week patterns provided additional information, while weather variables contributed less. This result is consistent with the strong autocorrelation observed during exploratory analysis.

Feature importance tells us which inputs the model leaned on most, not which inputs actually cause demand to change. It is also a different measurement from correlation. Correlation compares two variables directly, while feature importance only describes what the model did with the inputs we gave it.

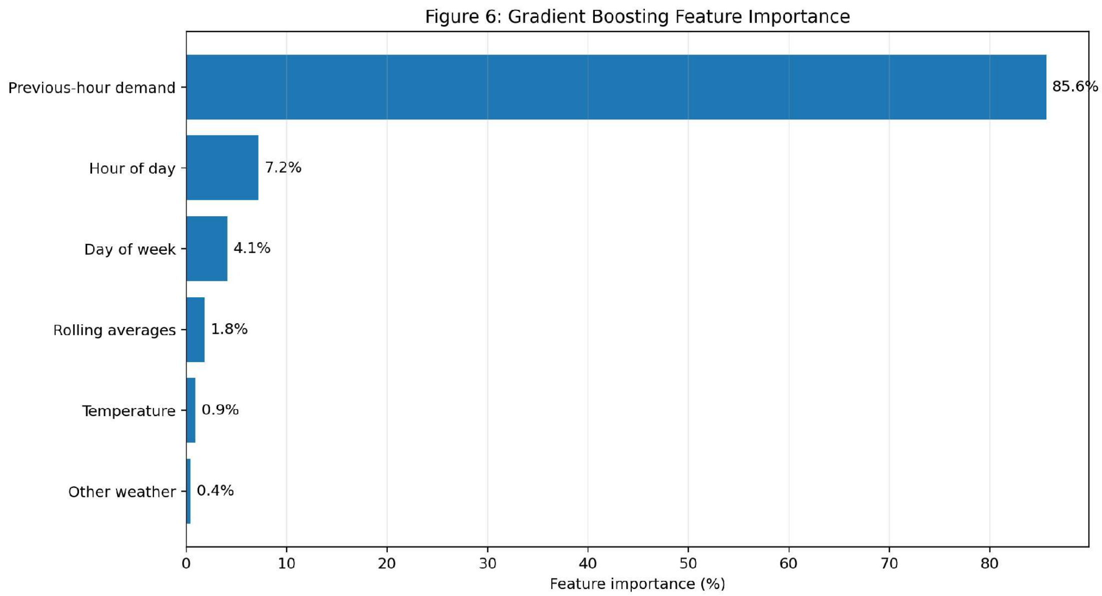

Figure 6. Gradient Boosting feature-importance analysis; previous-hour demand dominates the model.

5.4 Predicted vs. Actual Demand

The seven-day sample below shows that our Gradient Boosting predictions follow the main rises and falls in actual demand. It is a helpful picture to go with the error numbers, but one good week is not evidence that the model does equally well in every season or during extreme events.

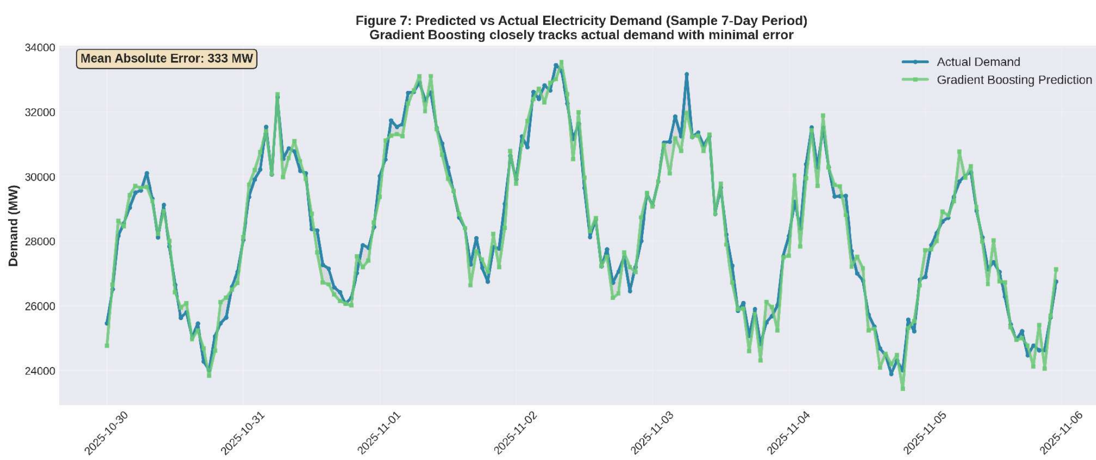

Figure 7. Gradient Boosting predictions compared with actual demand over a seven-day sample of the held-out test period. The sample MAE is 333 MW; the overall 60-day test MAE is 717 MW.

6. Discussion

6.1 Why Gradient Boosting Performed Well

One likely reason Gradient Boosting did well is that it can pick up combinations of inputs, like a hot weekday evening, that a straight-line model cannot represent. We gave it past demand, calendar patterns, rolling averages, and weather, so it could use the strong day-to-day repetition and then adjust it using the other information that a simple persistence forecast has no way to use.

Our results also point to something we did not expect: a more complicated model is not automatically better. The LSTM is designed to learn patterns over time, but it did not beat the simpler tree models here. We did not tune the LSTM much and our dataset is fairly small, so this is a result about the versions we tested rather than a ranking of forecasting methods in general.

6.2 Potential Energy-System Applications

Better short-term demand forecasts could help grid operators get generation and reserves ready, plan for high-demand periods, and match supply to expected load. They might also help with planning around solar and wind if they were combined with forecasts of renewable output. These are things our forecasts could be useful for, not things we measured.

We want to be clear that we did not measure any of those benefits. We measured forecasting accuracy only. We did not measure reserve levels, electricity costs, emissions, reliability, or how much renewable energy the grid could take. So our results suggest forecasting could be a useful input to grid operations, but they do not show that our model would reduce outages or costs.

6.3 Limitations and Future Work

- Geographic scope: the study uses one California grid region, so the findings may not generalize to other regions without retraining and testing.
- Time span: only one calendar year was modeled, limiting the ability to evaluate unusual multi-year changes and rare events.
- Weather information: the retrospective study uses observed historical weather rather than the weather forecast that would actually be available to an operator.
- Extreme events: rare grid emergencies and extreme weather conditions are underrepresented, so performance during those conditions remains uncertain.
- Model tuning: the LSTM was not tuned as extensively as the tree-based approaches, so the comparison should be interpreted as a comparison of the tested configurations.
- Testing which inputs matter: future work could train the models on past demand only, then add calendar features, then weather, to see how much each one actually helps.

A useful next step would be to repeat the experiment over several years and across multiple regions, while using operational weather forecasts rather than observed future weather. Another extension would be to evaluate forecast performance separately during normal periods, summer heat events, and other high-demand conditions.

7. Conclusion

This study examined whether machine-learning approaches could improve 24-hour-ahead electricity demand forecasts over a simple 24-hour persistence baseline. Using 2025 California electricity demand and historical weather data, the project compared five forecasting approaches—persistence, Ridge Regression, Random Forest, Gradient Boosting, and LSTM—using MAE, RMSE, and MAPE.

Among the configurations tested, Gradient Boosting performed best, achieving an MAE of 717 MW and a MAPE of 2.91%, compared with 1,049 MW and 4.28% for the persistence baseline. This represents a 31.7% reduction in MAE. Random Forest produced a similar result, while Ridge Regression and LSTM did not outperform the persistence baseline.

Our analysis also showed why forecasting demand is challenging and why persistence is a strong benchmark: demand has strong daily, weekly, and seasonal patterns, with a 24-hour autocorrelation of 0.913. Overall, the results suggest that tree-based machine learning can provide useful improvements for short-term electricity demand forecasting. Further research is needed to determine how well these findings generalize across years, regions, weather-forecast conditions, and extreme grid events.

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